

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

Education Analytics Dataset

```
data = {
    "Student_ID":
    ["S01", "S02", "S03", "S04", "S05", "S06", "S07", "S08", "S09", "S10",
    "S11", "S12", "S13", "S14", "S15", "S16", "S17", "S18", "S19", "S20"],
    "Attendance":
    [92, 55, 78, 60, 88, 47, 95, 72, 50, 83, 66, 90, 58, 76, 62, 85, 45, 80, 68, 53],
    "Study_Hours": [5, 2, 4, 3, 5, 1, 6, 3, 2, 4, 3, 5, 2, 4, 3, 5, 1, 4, 3, 2],
    "Internal_Marks":
    [45, 24, 38, 28, 42, 20, 48, 35, 22, 40, 30, 44, 25, 36, 29, 41, 18, 39, 32, 23],
    "Result":
    ["Pass", "Fail", "Pass", "Fail", "Pass", "Fail", "Pass", "Pass", "Fail", "Pass",
    ,
    "Fail", "Pass", "Fail", "Pass", "Fail", "Pass", "Fail", "Pass", "Pass", "Fail"]
}

df = pd.DataFrame(data)

print(df)
```

	Student_ID	Attendance	Study_Hours	Internal_Marks	Result
0	S01	92	5	45	Pass
1	S02	55	2	24	Fail
2	S03	78	4	38	Pass
3	S04	60	3	28	Fail
4	S05	88	5	42	Pass
5	S06	47	1	20	Fail
6	S07	95	6	48	Pass
7	S08	72	3	35	Pass
8	S09	50	2	22	Fail
9	S10	83	4	40	Pass
10	S11	66	3	30	Fail
11	S12	90	5	44	Pass
12	S13	58	2	25	Fail
13	S14	76	4	36	Pass
14	S15	62	3	29	Fail
15	S16	85	5	41	Pass
16	S17	45	1	18	Fail
17	S18	80	4	39	Pass
18	S19	68	3	32	Pass
19	S20	53	2	23	Fail

```
print(df.duplicated().sum())
```

```
0
```

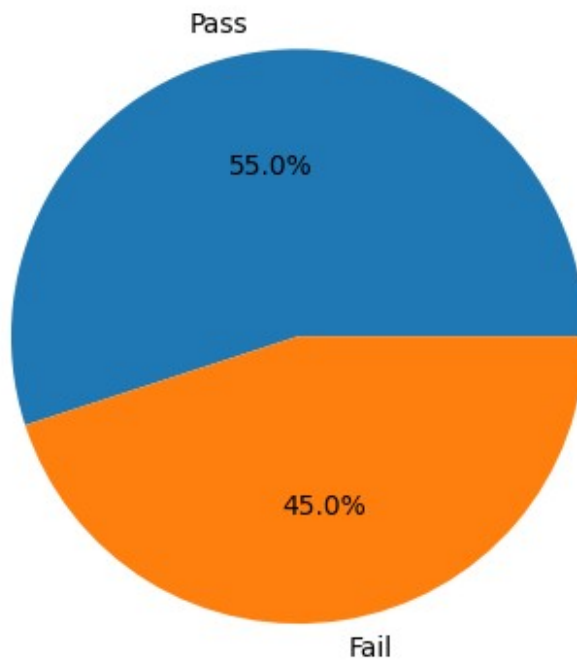
```
print(df.describe)
```

```
<bound method NDFrame.describe of
Study_Hours  Internal_Marks  Result  Student_ID  Attendance
0           S01           92         5           45      Pass
1           S02           55         2           24      Fail
2           S03           78         4           38      Pass
3           S04           60         3           28      Fail
4           S05           88         5           42      Pass
5           S06           47         1           20      Fail
6           S07           95         6           48      Pass
7           S08           72         3           35      Pass
8           S09           50         2           22      Fail
9           S10           83         4           40      Pass
10          S11           66         3           30      Fail
11          S12           90         5           44      Pass
12          S13           58         2           25      Fail
13          S14           76         4           36      Pass
14          S15           62         3           29      Fail
15          S16           85         5           41      Pass
16          S17           45         1           18      Fail
17          S18           80         4           39      Pass
18          S19           68         3           32      Pass
19          S20           53         2           23      Fail>
```

```
counts = df["Result"].value_counts()
```

```
plt.pie(counts, labels=counts.index, autopct='%1.1f%%')
```

```
plt.show()
```



```
df2=pd.read_csv("student_data.csv")
```

```
print(df2)
```

	school	sex	age	address	famsize	Pstatus	Medu	Fedu	Mjob
0	GP	F	18	U	GT3	A	4	4	at_home
1	GP	F	17	U	GT3	T	1	1	at_home
2	GP	F	15	U	LE3	T	1	1	at_home
3	GP	F	15	U	GT3	T	4	2	health
4	GP	F	16	U	GT3	T	3	3	other
...	...	...	...	...	...	...	...	...	...
390	MS	M	20	U	LE3	A	2	2	services
391	MS	M	17	U	LE3	T	3	1	services
392	MS	M	21	R	GT3	T	1	1	other
393	MS	M	18	R	LE3	T	3	2	services
394	MS	M	19	U	LE3	T	1	1	other

at_home

	...	famrel	freetime	goout	Dalc	Walc	health	absences	G1	G2
G3										
0	...	4	3	4	1	1	3	6	5	6
6										
1	...	5	3	3	1	1	3	4	5	5
6										
2	...	4	3	2	2	3	3	10	7	8
10										
3	...	3	2	2	1	1	5	2	15	14
15										
4	...	4	3	2	1	2	5	4	6	10
10										
..	...	...	...	...	...	...	...	...	..	..
.										
390	...	5	5	4	4	5	4	11	9	9
9										
391	...	2	4	5	3	4	2	3	14	16
16										
392	...	5	5	3	3	3	3	3	10	8
7										
393	...	4	4	1	3	4	5	0	11	12
10										
394	...	3	2	3	3	3	5	5	8	9
9										

[395 rows x 33 columns]

print(df2.info)

```
<bound method DataFrame.info of
Pstatus  Medu  Fedu      Mjob      Fjob  school sex  age address famsize
0      GP   F   18      U   GT3      A    4    4    at_home
teacher
1      GP   F   17      U   GT3      T    1    1    at_home
other
2      GP   F   15      U   LE3      T    1    1    at_home
other
3      GP   F   15      U   GT3      T    4    2    health
services
4      GP   F   16      U   GT3      T    3    3    other
other
..      ..   ..   ..      ..      ..      ..      ..      ..
...
390     MS   M   20      U   LE3      A    2    2    services
services
391     MS   M   17      U   LE3      T    3    1    services
services
392     MS   M   21      R   GT3      T    1    1    other
```

```

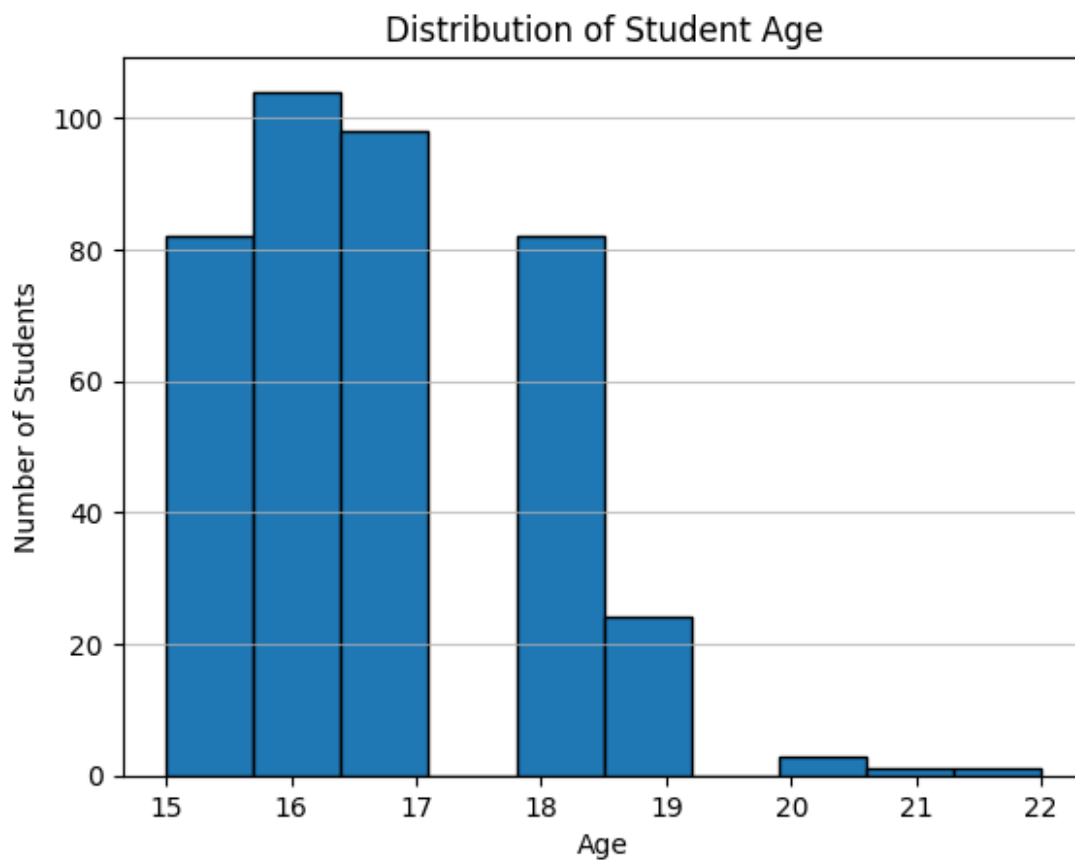
other
393      MS    M    18      R    LE3      T    3    2  services
other
394      MS    M    19      U    LE3      T    1    1    other
at_home

... famrel freetime goout Dalc Walc health absences G1 G2
G3
0      ...      4      3      4      1      1      3      6      5      6
6
1      ...      5      3      3      1      1      3      4      5      5
6
2      ...      4      3      2      2      3      3     10      7      8
10
3      ...      3      2      2      1      1      5      2     15     14
15
4      ...      4      3      2      1      2      5      4      6     10
10
..      ...      ...      ...      ...      ...      ...      ...      ..      ..      .
.
390     ...      5      5      4      4      5      4     11      9      9
9
391     ...      2      4      5      3      4      2      3     14     16
16
392     ...      5      5      3      3      3      3      3     10      8
7
393     ...      4      4      1      3      4      5      0     11     12
10
394     ...      3      2      3      3      3      5      5      8      9
9

[395 rows x 33 columns]>

plt.hist(df2['age'], bins=10, edgecolor='black')
plt.title('Distribution of Student Age')
plt.xlabel('Age')
plt.ylabel('Number of Students')
plt.grid(axis='y', alpha=0.75)
plt.show()

```



Healthcare Dataset – EDA

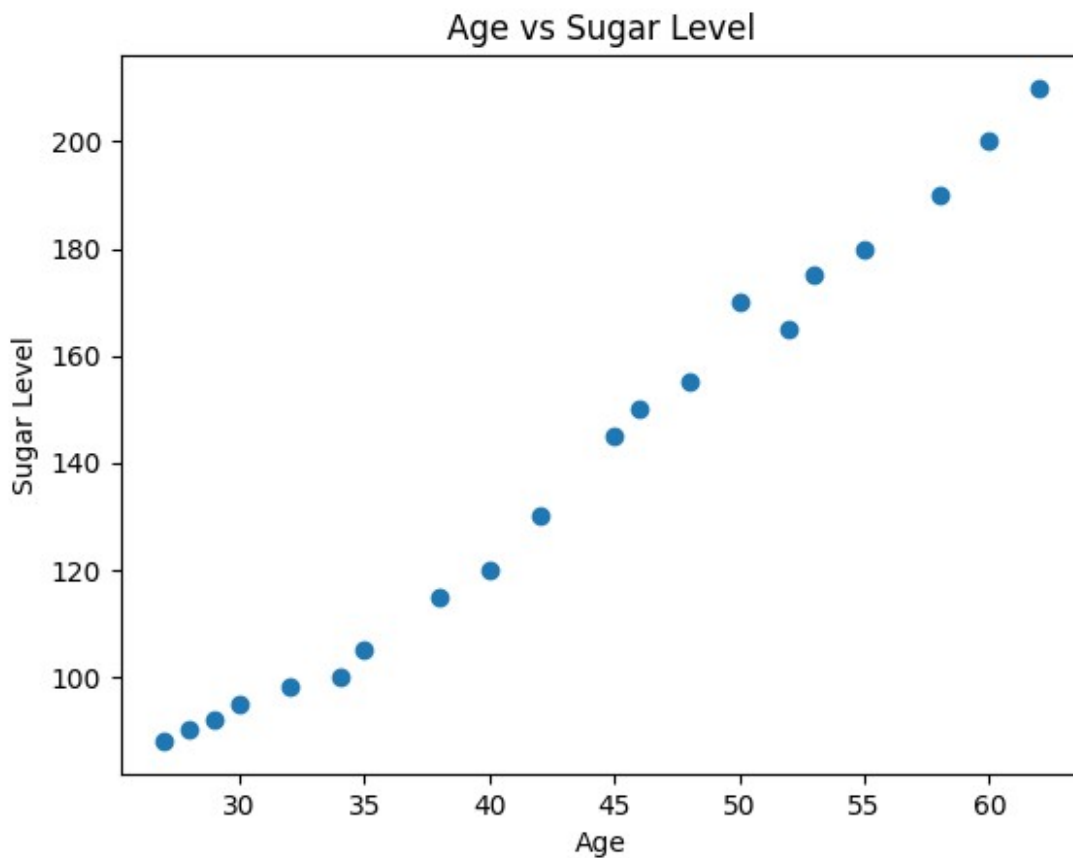
```
data = {
    "Patient_ID":
    ["P01", "P02", "P03", "P04", "P05", "P06", "P07", "P08", "P09", "P10",
    "P11", "P12", "P13", "P14", "P15", "P16", "P17", "P18", "P19", "P20"],
    "Age":
    [45, 32, 55, 28, 60, 40, 52, 35, 48, 30, 62, 38, 50, 27, 58, 42, 46, 34, 53, 29],
    "Sugar_Level":
    [145, 98, 180, 90, 200, 120, 165, 105, 155, 95, 210, 115, 170, 88, 190, 130, 150, 100, 175, 92],
    "BP":
    [130, 118, 145, 110, 150, 125, 140, 120, 135, 115, 155, 122, 142, 108, 148, 128, 132, 116, 144, 112],
    "BMI":
    [28, 23, 31, 22, 34, 26, 30, 24, 29, 23, 35, 25, 30, 21, 33, 27, 28, 24, 31, 22],
    "Disease_Status":
    ["Yes", "No", "Yes", "No", "Yes", "No", "Yes", "No", "Yes", "No",
    "Yes", "No", "Yes", "No", "Yes", "No", "Yes", "No", "Yes", "No"]
}
```

```
df = pd.DataFrame(data)
```

```
print(df.describe())
```

	Age	Sugar_Level	BP	BMI
count	20.000000	20.000000	20.000000	20.000000
mean	43.200000	138.65000	129.750000	27.300000
std	11.316499	40.33449	14.523574	4.256265
min	27.000000	88.00000	108.000000	21.000000
25%	33.500000	99.50000	117.500000	23.750000
50%	43.500000	137.50000	129.000000	27.500000
75%	52.250000	171.25000	142.500000	30.250000
max	62.000000	210.00000	155.000000	35.000000

```
plt.scatter(df["Age"], df["Sugar_Level"])  
plt.xlabel("Age")  
plt.ylabel("Sugar Level")  
plt.title("Age vs Sugar Level")  
plt.show()
```



Retail Sales Dataset

```

data = {
    "Product_ID":
["PR01", "PR02", "PR03", "PR04", "PR05", "PR06", "PR07", "PR08", "PR09", "PR10"
,
"PR11", "PR12", "PR13", "PR14", "PR15", "PR16", "PR17", "PR18", "PR19", "PR20"]
,
    "Category":
["Grocery", "Snacks", "Dairy", "Fruits", "Vegetables", "Grocery", "Snacks", "Dairy", "Fruits", "Vegetables",
"Grocery", "Snacks", "Dairy", "Fruits", "Vegetables", "Grocery", "Snacks", "Dairy", "Fruits", "Vegetables"],
    "Price":
[50, 30, 45, 60, 40, 80, 25, 55, 70, 35, 90, 20, 65, 75, 45, 100, 35, 50, 80, 30],
    "Quantity_Sold":
[20, 35, 18, 25, 30, 15, 40, 16, 22, 28, 12, 45, 14, 20, 26, 10, 32, 19, 18, 35],
    "Discount": [5, 3, 2, 5, 4, 6, 3, 2, 5, 4, 6, 2, 3, 5, 4, 7, 3, 2, 6, 3],
    "Revenue":
[1000, 1050, 810, 1500, 1200, 1200, 1000, 880, 1540, 980, 1080, 900, 910, 1500, 1170, 1000, 1120, 950, 1440, 1050]
}

```

```
df = pd.DataFrame(data)
```

```
print(df.head())
```

	Product_ID	Category	Price	Quantity_Sold	Discount	Revenue
0	PR01	Grocery	50	20	5	1000
1	PR02	Snacks	30	35	3	1050
2	PR03	Dairy	45	18	2	810
3	PR04	Fruits	60	25	5	1500
4	PR05	Vegetables	40	30	4	1200

```
print(df.info())
```

```
print(df.describe())
```

```
print(df.isnull().sum())
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20 entries, 0 to 19
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Product_ID      20 non-null    object
1   Category        20 non-null    object
2   Price           20 non-null    int64

```



```
3   Quantity_Sold    20 non-null    int64
4   Discount         20 non-null    int64
5   Revenue          20 non-null    int64
```

```
dtypes: int64(4), object(2)
```

```
memory usage: 1.1+ KB
```

```
None
```

	Price	Quantity_Sold	Discount	Revenue
count	20.000000	20.000000	20.000000	20.000000
mean	54.000000	24.000000	4.000000	1114.000000
std	22.803509	9.673621	1.555973	221.416493
min	20.000000	10.000000	2.000000	810.000000
25%	35.000000	17.500000	3.000000	972.500000
50%	50.000000	21.000000	4.000000	1050.000000
75%	71.250000	30.500000	5.000000	1200.000000
max	100.000000	45.000000	7.000000	1540.000000

```
Product_ID    0
```

```
Category      0
```

```
Price         0
```

```
Quantity_Sold 0
```

```
Discount      0
```

```
Revenue       0
```

```
dtype: int64
```

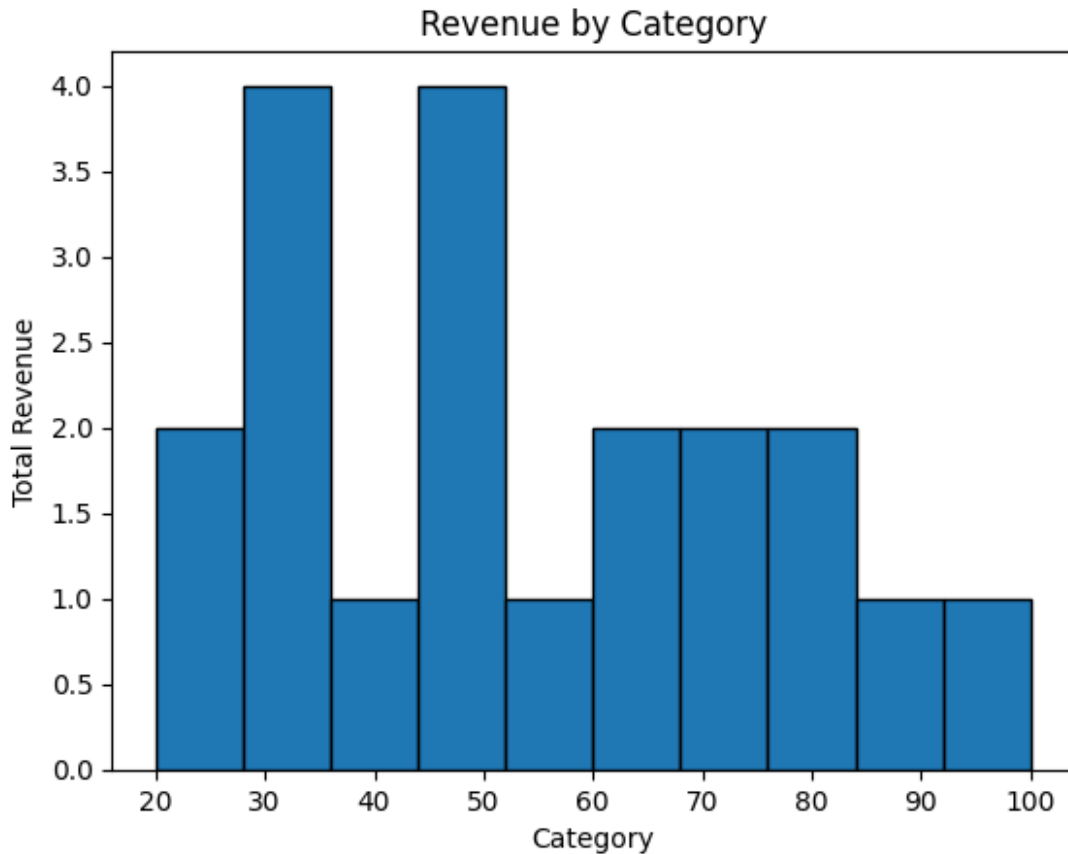
```
plt.hist(df['Price'], bins=10, edgecolor='black')
```

```
plt.title("Revenue by Category")
```

```
plt.xlabel("Category")
```

```
plt.ylabel("Total Revenue")
```

```
plt.show()
```



Banking Loan Approval

```
data = {  
    "Customer_ID":  
    ["C01", "C02", "C03", "C04", "C05", "C06", "C07", "C08", "C09", "C10",  
    "C11", "C12", "C13", "C14", "C15", "C16", "C17", "C18", "C19", "C20"],  
    "Income":  
    [45000, 25000, 60000, 22000, 50000, 30000, 70000, 28000, 55000, 24000,  
    65000, 35000, 48000, 26000, 75000, 32000, 52000, 27000, 68000, 23000],  
    "Credit_Score": [720, 580, 750, 560, 710, 600, 780, 590, 730, 570,  
    760, 620, 700, 585, 790, 610, 725, 575, 770, 555],  
    "Loan_Amount":  
    [200000, 150000, 300000, 120000, 250000, 180000, 350000, 160000, 270000, 130000,  
    ,  
    320000, 200000, 240000, 140000, 400000, 190000, 260000, 150000, 330000, 125000]  
    ,  
    "Employment_Type": ["Salaried", "Self-  
employed", "Salaried", "Unemployed", "Salaried", "Self-employed",  
    "Salaried", "Self-  
employed", "Salaried", "Unemployed", "Salaried", "Self-employed",
```

```

        "Salaried", "Unemployed", "Salaried", "Self-
employed", "Salaried", "Self-employed",
        "Salaried", "Unemployed"],
    "Loan_Status":
["Approved", "Rejected", "Approved", "Rejected", "Approved", "Rejected", "Ap
proved", "Rejected",
"Approved", "Rejected", "Approved", "Rejected", "Approved", "Rejected", "App
proved", "Rejected",
    "Approved", "Rejected", "Approved", "Rejected"]
}

```

```

df = pd.DataFrame(data)
print(df)

```

	Customer_ID	Income	Credit_Score	Loan_Amount	Employment_Type	Loan_Status
0	C01	45000	720	200000	Salaried	Approved
1	C02	25000	580	150000	Self-employed	Rejected
2	C03	60000	750	300000	Salaried	Approved
3	C04	22000	560	120000	Unemployed	Rejected
4	C05	50000	710	250000	Salaried	Approved
5	C06	30000	600	180000	Self-employed	Rejected
6	C07	70000	780	350000	Salaried	Approved
7	C08	28000	590	160000	Self-employed	Rejected
8	C09	55000	730	270000	Salaried	Approved
9	C10	24000	570	130000	Unemployed	Rejected
10	C11	65000	760	320000	Salaried	Approved
11	C12	35000	620	200000	Self-employed	Rejected
12	C13	48000	700	240000	Salaried	Approved
13	C14	26000	585	140000	Unemployed	Rejected
14	C15	75000	790	400000	Salaried	Approved
15	C16	32000	610	190000	Self-employed	Rejected
16	C17	52000	725	260000	Salaried	

Approved					
17	C18	27000	575	150000	Self-employed
Rejected					
18	C19	68000	770	330000	Salaried
Approved					
19	C20	23000	555	125000	Unemployed
Rejected					

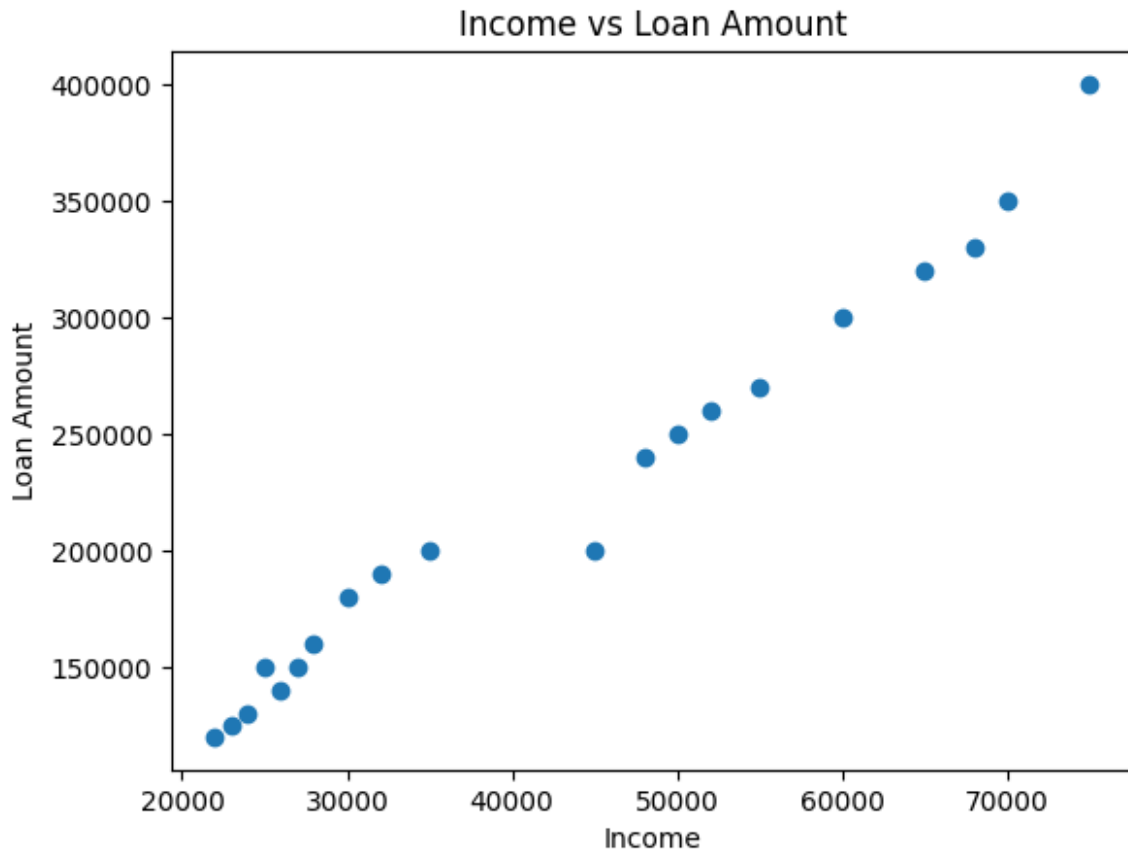
```
print(df.duplicated().sum())
```

```
0
```

```
print(df.isnull().sum())
```

```
Customer_ID      0
Income           0
Credit_Score     0
Loan_Amount      0
Employment_Type  0
Loan_Status      0
dtype: int64
```

```
plt.scatter(df["Income"], df["Loan_Amount"])
plt.title("Income vs Loan Amount")
plt.xlabel("Income")
plt.ylabel("Loan Amount")
plt.show()
```



Agriculture Crop Yield

```
data2 = {  
    "Farm_ID":  
    ["F01", "F02", "F03", "F04", "F05", "F06", "F07", "F08", "F09", "F10",  
    "F11", "F12", "F13", "F14", "F15", "F16", "F17", "F18", "F19", "F20"],  
    "Rainfall_mm":  
    [850, 600, 900, 500, 780, 650, 920, 720, 550, 800, 880, 620, 760, 940, 580, 820, 700, 960, 540, 790],  
    "Temperature":  
    [28, 32, 27, 34, 29, 31, 26, 30, 33, 28, 27, 32, 29, 26, 33, 28, 30, 25, 34, 29],  
    "Fertilizer_kg":  
    [60, 45, 65, 40, 55, 48, 70, 52, 42, 58, 62, 46, 54, 72, 44, 60, 50, 75, 41, 56],  
    "Soil_Type":  
    ["Loamy", "Sandy", "Loamy", "Sandy", "Clay", "Sandy", "Loamy", "Clay", "Sandy",  
    "Loamy",  
    "Loamy", "Sandy", "Clay", "Loamy", "Sandy", "Clay", "Clay", "Loamy", "Sandy", "  
    Clay"],  
    "Crop_Yield_kg":  
    [3200, 2100, 3500, 1800, 3000, 2300, 3700, 2800, 1900, 3100,
```

```
3400,2200,2950,3800,2000,3150,2700,3900,1850,3050]
}
```

```
df2 = pd.DataFrame(data2)
```

```
print(df2.describe())
```

	Rainfall_mm	Temperature	Fertilizer_kg	Crop_Yield_kg
count	20.000000	20.000000	20.000000	20.000000
mean	743.000000	29.550000	54.750000	2820.000000
std	144.335136	2.762055	10.507516	681.407213
min	500.000000	25.000000	40.000000	1800.000000
25%	615.000000	27.750000	45.750000	2175.000000
50%	770.000000	29.000000	54.500000	2975.000000
75%	857.500000	32.000000	60.500000	3250.000000
max	960.000000	34.000000	75.000000	3900.000000

```
df2.boxplot(column="Crop_Yield_kg", by="Soil_Type")
plt.title("Yield by Soil Type")
plt.show()
```

